

ASSIGNMENT 2 - REPORT

Building Detection from Satellite Imagery using YOLOv8

1. Problem

The objective of this project is to automatically detect and count buildings from satellite imagery using a deep learning object detection model. Building detection is an important task in urban planning, disaster management, population estimation, infrastructure monitoring, and mapping applications. The final system takes a satellite image as input and outputs the detected building locations along with the total building count.

2. Data

Dataset Source

The dataset used for this project is the SpaceNet Building Detection Dataset obtained from Kaggle.

Dataset Description

The dataset consists of satellite image tiles and corresponding annotation files. Each annotation file contains bounding box coordinates representing the locations of buildings in the image.

Dataset Statistics

- Total Images: 1148
- Total Label Files: 1148
- Training Images: 918
- Validation Images: 115
- Test Images: 115

The dataset was split using an 80/10/10 ratio for training, validation, and testing.

Label Format

Annotations are stored in YOLO format:

`class_id x_center y_center width height`

where all coordinates are normalized between 0 and 1.

Dataset Analysis

The average number of buildings per image was found to be:

14.49 buildings/image

Most satellite images have consistent resolution and image quality. However, the density of buildings varies significantly from image to image.

Challenges

Some expected challenges include:

- **Dense clusters of buildings causing overlapping detections.**
 - **Small rooftops that occupy only a few pixels.**
 - **Shadows from tall structures.**
 - **Variations in rooftop color and shape.**
 - **Background objects that may resemble buildings.**
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3. Model

What is YOLOv8?

YOLOv8 (You Only Look Once Version 8) is a state-of-the-art object detection framework developed by Ultralytics. It performs object localization and classification in a single forward pass, making it both accurate and computationally efficient.

Model Selection

The YOLOv8 Small model (YOLOv8s) was selected for this project.

Instead of training from random initialization, pretrained weights (yolov8s.pt) were used. These weights already contain useful visual features learned from large-scale datasets, allowing faster convergence and better performance on custom datasets through transfer learning.

Model Information

- **Model: YOLOv8s**
 - **Parameters: 11,125,971**
 - **Classes: 1**
 - **Class Name: Building**
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4. Training

Training Configuration

Parameter	Value
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Model	YOLOv8s
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Epochs	25
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Image Size	640 × 640
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Batch Size	16
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Classes	1
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GPU	NVIDIA Tesla T4
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Training Results

The final training results obtained from results.csv were:

Metric	Value
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Final Box Loss	1.229
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Final mAP@50	0.793
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Training Curve Analysis

The training and validation losses decreased steadily during training, indicating successful learning of building features.

The mAP@50 score improved consistently throughout training and began to stabilize near the final epochs, suggesting that the model was approaching convergence. There was no strong evidence of severe overfitting because validation performance remained close to training performance.

Insert Train vs Validation Loss Plot Here

Insert mAP@50 Curve Here

5. Evaluation

The trained model was evaluated on the validation/test data using Ultralytics evaluation tools.

Performance Metrics

Metric	Value	Meaning
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mAP@50	0.794	Overall detection accuracy at IoU threshold 0.50
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Precision	0.790	About 79% of predicted buildings were correct
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Recall	0.734	About 73% of actual buildings were detected
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F1 Score	0.761	Balanced measure of precision and recall
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Interpretation

The model achieved a strong mAP@50 score of approximately 79.4%, indicating good localization and detection performance.

A precision of 79% shows that false detections were relatively limited, while a recall of 73.4% indicates that most buildings present in the images were successfully detected.

Insert Precision–Recall Curve Here

Insert Confusion Matrix Here

6. Building Detection Pipeline

A reusable detection pipeline was developed using the trained YOLOv8 model.

The pipeline performs the following steps:

1. Load input image.
2. Run YOLOv8 inference.
3. Detect building bounding boxes.
4. Count detected buildings.
5. Calculate average confidence.
6. Generate annotated output image.
7. Report processing time.

Counting Accuracy

The building counting performance was tested on 10 images from the test set.

Mean Absolute Error (MAE) = 2.3

This means the predicted building count differed from the true count by approximately 2.3 buildings on average.

Example Prediction

Example output from the notebook:

- Image: AOI_3_Paris_img52.png
- Buildings Detected: 0
- Average Confidence: 0.00
- Processing Time: 33.84 ms

Insert Sample Prediction Image Here

7. Observations

What Worked Well

- Transfer learning enabled fast convergence.
- YOLOv8 successfully detected most medium and large buildings.
- The model achieved a strong mAP@50 score close to 80%.
- Inference was fast, making real-time deployment feasible.

Difficult Cases

- Small buildings were sometimes missed.
- Closely packed buildings occasionally merged into a single detection.
- Shadows and low-contrast rooftops reduced detection accuracy.
- Buildings with unusual shapes were harder to identify.

Possible Improvements

Future improvements may include:

1. Training for more epochs.
2. Using larger YOLOv8 variants such as YOLOv8m or YOLOv8l.
3. Applying stronger data augmentation techniques.
4. Increasing input image resolution.
5. Using additional satellite imagery from different geographical regions.

8. Conclusion

In this project, a complete building detection and counting system was developed using YOLOv8 and satellite imagery from the SpaceNet dataset. The trained model achieved an mAP@50 of 79.4%, Precision of 79.0%, Recall of 73.4%, and an F1 Score of 76.1%. The system successfully detects and counts buildings while maintaining fast inference speed, demonstrating the effectiveness of modern object detection techniques for remote sensing applications.